Lecture 2: Introduction to Limits

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Day 1 Recap

Questions:

1. Was everyone able to access the Gradescope/homework?
2. How did the homework go? (Note: Length will not be as long for future HWs, this one just had a lot of topics to review)
3. Any questions, comments, concerns?

Quiz Time!

At this point it's quiz time!

1. Get into neatly organized rows
2. Do not flip over the page and start until directed to
3. Should go without saying but no cheating!
    - No looking at neighbors papers
    - No calculators
    - No notes
    - No phones
    - etc. etc.
4. I'll write on the board the minutes you have left, crossing each number out when you have *less than* that number of minutes left to complete
5. When time is up, stop writing and hold your quiz up so I can grab it. If you finish early, just chill until the 5 mins are up.

Time to do some math!

Now let’s start the lecture!

Limits

MATH 1271 - Part 1: Limits

We will start our course by discussing the concept of a *limit of a function*.
Intuitively, the limit serves several important purposes:

- It gives you information about the value that a function tends towards, as it's input tends towards some fixed input.
- It tells you what value the function is "approximating" around some fixed input.
- It tells you local information about the function around some *small neighborhood* of some fixed input.

Approximation I

Let's run with the second interpretation for a bit:
Say you're doing some experiment where you're studying the relationship between some *input variable x* (maybe it's the concentration of some acid in a solution) and some *output variable y* (maybe it's the pH of that solution).
Real world problems:

1. We can almost never perfectly measure physical quantities e.g.
   (1) We can't know the exact number of molecules of an acid present in a solution, only up to some error
   (2) Using typical measurements like pH strips etc. we also can't get a precise measurement of the output variable either

How do we proceed if, for instance, we want to know the *exact* pH of a solution whose concentration of acid is *exactly* 0.5 moles per liter?

Approximation II

We can either:

1. Make many many many measurements and take their average, or
2. Use more precise methods of measurements that decrease the error range as much as possible.

Graphically:

Approximation III

WARNING!

How do we actually know that this average we're taking is the true output value?

What it really is, is...
The Limit!

Limits I

Limit of a Function

For some function $f : \mathbb{R} \to \mathbb{R}$ and some fixed input $x_0 \in \mathbb{R}$, we say that $\lim_{x \to x_0} f(x) = L$ (said "the limit of $f(x)$ as x approaches x_0 is L") if we can make $f(x)$ as close to L as we want just by forcing x to be close enough to x_0.

Explicitly: for any error tolerance $\epsilon > 0$, if we want $f(x)$ to be equal to L within an error of ϵ (e.g. $|f(x) - L| < \epsilon$), then we just need to find some input error tolerance $\delta > 0$ and ensure that x is equal to x_0 within an error of δ (e.g. $|x - x_0| < \delta$).

If no such value L exists (i.e. the function *cannot* be well approximated around x_0) then we say that $\lim_{x \to x_0} f(x)$ *does not exist*.

Limits II

How can we find limits?
Option 1 (easiest): Graphically!
Given a graph of a function, we can easily tell what value a function approaches just by examining where the graph tends towards as we near a given point.

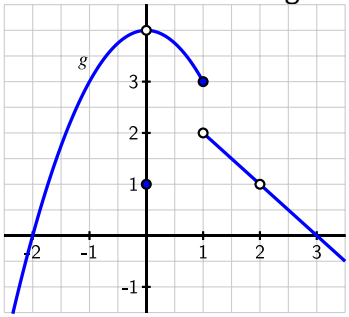


Figure: An example function

Limits III

What if we don't have a graph to work with?

Option 2 (medium easy, *only an estimate!*): Evaluation!

Given an expression of a function, e.g. $f(x) = x^2$ and a specific value around which to compute the limit, e.g. $x_0 = 2$, we can compute values that are closer and closer to x_0 *by hand/calculator* and guess if they're approaching a value. (Similar to the method we talked about in the approximation section.)

Limits IV

What about if we are both:

1. Not given a graph
2. Interested in the *exact* limit, not an approximation?

Option 3 (exact!): Algebraic Manipulation!
E.g., if we wanted to find $\lim_{x \to 2} x^2$ we could:

1. Rewrite it as $\lim_{h \to 0} (2 + h)^2$
2. Expand the argument to get $\lim_{h \to 0} 4 + 2h + h^2$
3. Group the h terms together to get $\lim_{h \to 0} 4 + h(2 + h)$
4. Notice that as $h \to 0$, $h(2 + h) \to 0$ as well so $\lim_{x \to 2} x^2 = 4$

Limits V

Notice that $\lim_{x \to 2} x^2 = 4 = 2^2 = f(2)$ i.e. *the limit of* $f(x)$ *as* $x \to x_0$ *is equal to* $f(x_0)$!
Is this always true? I.e. no matter what function f we choose and what fixed input x_0 we take, do we always have that:

$$\lim_{x \to x_0} f(x) \stackrel{?}{=} f(x_0) \tag{1}$$

Limits VI

NO!

Two things can go wrong:

1. We can artifically redefine the value of a function as much as we want:

$$f(x) = \begin{cases} 4, & x = 0 \\ x^2, & x \neq 0 \end{cases} \tag{2}$$

2. We can have a function that isn't *defined* at a point but still *has a limit at that point*:

$$g(x) = \frac{x^2 - 1}{x + 1}. \tag{3}$$

Limits VII

Let's study that $g(x) = \dfrac{x^2 - 1}{x + 1}$ function more:

1. What is the domain of g?
2. For points x_0 in the domain of g, is it true that $\lim_{x \to x_0} g(x) = g(x_0)$?
3. What about the point that is *not* in the domain of g? What is $\lim_{x \to -1} g(x)$?

Limits VIII

Let's try another weird function:

$$h(x) = \sin(\frac{\pi}{x}). \tag{4}$$

1. Domain?
2. Does limit=value for points in the domain?
3. What about $x_0 = 0$?

Instantaneous Velocity I

What can we *do* with limits?

Average Velocity

Suppose that an object's position as a function of time t is given by the function $s(t)$.
The average velocity of the object, from time $t = a$ to time $t = b$ is given by:

$$AV_{[a,b]} := \frac{s(b) - s(a)}{b - a}. \tag{5}$$

However, we rarely ever see *average* velocity in real life! E.g. if you drive a car then the dashboard reports your speed exactly in the instant you look at it, not over the last 1, 2, 5, etc. minutes!
(If it did, imagine someone driving slowly for a while, rapidly accelerating to a dangerous speed, getting pulled over and then trying to tell the cop "oh well my car said my average speed over the past 5 minutes was just 20mph" unlikely to avoid a ticket!)

Instantaneous Velocity II

That why we care about the *instantaneous* velocity of an object:

Instantaneous Velocity

Suppose that an object's position as a function of time t is given by the function $s(t)$. The instantaneous velocity of the object at time $t = a$ is defined as

$$IV_{t=a} = \lim_{\Delta t \to 0} AV_{[a,a+\Delta t]} = \lim_{h \to 0} \frac{s(a+h) - s(a)}{h} \tag{6}$$

Instantaneous Velocity III

Suppose that $s(t) = \sqrt{t+1}$. What is the instantaneous velocity at $t = 0$?

1. What is the domain of $\dfrac{s(0+h) - s(0)}{h} = \dfrac{\sqrt{1+h} - 1}{h}$?
2. Can we determine the limit just by evaluating the above function at $h = 0$?
3. How could we get $\dfrac{\sqrt{1+h} - 1}{h}$ into a form where we *could* just evaluate?

Worksheet Time!

Now let's get into groups and work on the worksheet!

End of Class!

That’s all folks!